

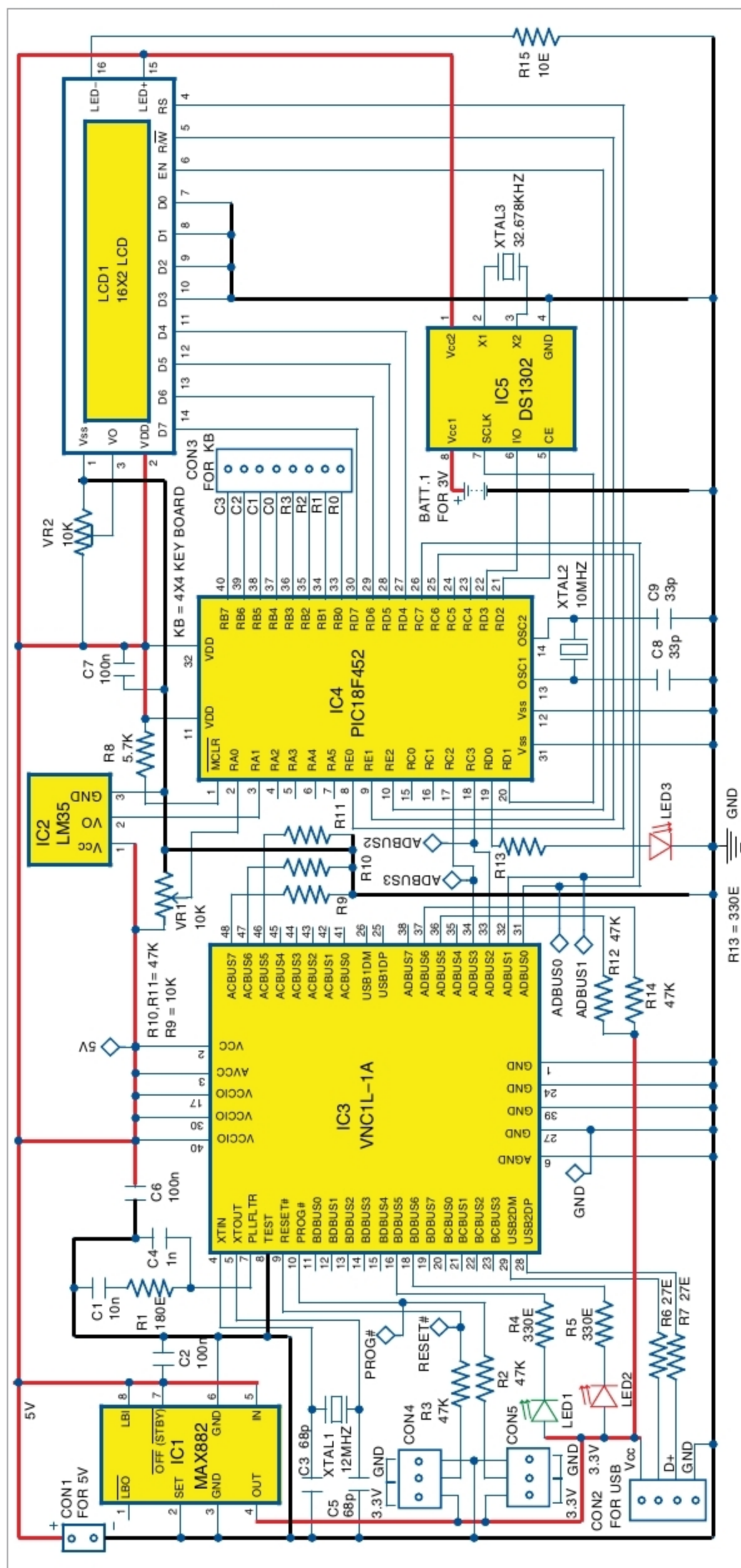


Fig. 3: Circuit diagram of the real-time USB data logger

Software description

A firmware program for sampling two analogue voltages applied on pins 2 and 3 of PIC18F452 at regular intervals of time and storing data in a file in the USB flash drive attached to VNC1L-1A is explained here. The firmware has been written using CCS C compiler and is uploaded into PIC18F452 flash memory.

The firmware includes modules performing the following tasks:

Synchronising UART communication. UART communication between VNC1L-1A and PIC18F452 is ensured by receiving echoes of E and e characters sent from PIC18F452.

Receive Data Available interrupt (INT_RDA) handling. The interrupt is generated in PIC18F452 whenever a byte is received from VNC1L-1A in receive buffer of PIC18F452 via UART interface. The program reads the character and sends to USB flash drive.

Sampling analogue channels. The firmware includes codes to sample analogue channels 0 and 1. Also, CCS C Compiler has inbuilt functions to configure the on-chip ADC for selecting channels and acquiring converted data. These functions can be used by the program for sampling analogue channels.

Codes to write data into USB flash drive. The firmware includes codes to send commands to create Results.dat file in the flash drive, to write twenty values of sampled data into the file and to close the file.

RTC and user interface

In the present design, a user interface is provided with the keyboard for input and LCD for output display to monitor time duration. DS1302 chip provides real-time data to the system. Interfacing DS1302 with an MCU is simplified by using synchronous serial communication. Only three wires are needed to communicate with DS1302 for chip enable (CE), I/O (data line) and serial clock (SCLK) signals.

During read or write operation, CE must be asserted high since this pin has an internal 40kΩ pull-down resistor to ground. SCLK is used to synchronise data movement in the serial interface. Moreover, DS1302 is designed to operate on very low power such that it retains data and clocks information with a power consump-